THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · NCERT-TYPE UNIT TEST SET 3

Chapters 1–4: **Exploring Substances: Acidic, Basic and Neutral, Electricity: Circuits and their Components, The World of Metals and Non-metals, Changes Around Us: Physical and Chemical** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Which non-metal is a liquid at room temperature? 1

(a) sulphur **(b)** bromine **(c)** carbon **(d)** iodine

2. Which of the following is a chemical change? 1

(a) stretching a rubber band **(b)** burning of magnesium ribbon

(c) breaking a glass **(d)** cutting hair

3. Turmeric indicator turns red in: 1

(a) neutral **(b)** acid **(c)** base **(d)** vinegar

4. An electric heater has a coil of: 1

(a) plastic **(b)** wood **(c)** nichrome wire **(d)** copper wire

5. Non-metal oxides are generally: 1

(a) basic **(b)** neutral **(c)** soapy **(d)** acidic

6. Rusting can be prevented by: 1

(a) exposing to acid **(b)** keeping in moist air

(c) painting **(d)** leaving in water

7. Which of these is basic? 1

(a) curd **(b)** baking soda solution

(c) lemon juice **(d)** vinegar

8. A circuit in which the current flows continuously is: 1

(a) a closed circuit **(b)** an open circuit **(c)** a short wire **(d)** a broken circuit

9. Metals produce a ringing sound when struck. This property is: 1

(a) ductility **(b)** malleability **(c)** lustre **(d)** sonority

10. Burning of paper is a: 1

(a) chemical change **(b)** change of state

(c) physical change **(d)** reversible change

11. The formula of calcium hydroxide (slaked lime) is: 1

(a) CaO **(b)** $Ca(OH)_2$ **(c)** $CaCO_3$ **(d)** $NaOH$

12. Which of these is an insulator? 1

(a) plastic **(b)** aluminium **(c)** graphite **(d)** iron

13. Which non-metal is lustrous? 1

(a) carbon (coal) **(b)** iodine **(c)** sulphur **(d)** phosphorus

14. Which of the following is a physical change? 1

(a) stretching a rubber band **(b)** digesting food

(c) burning wood **(d)** rusting nails

15. China rose indicator turns dark pink in: 1

(a) acid **(b)** water **(c)** neutral **(d)** base

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Why is graphite used in electrodes and pencils? 2

17. What happens when baking soda is added to vinegar? 2

18. What is the effect of acids and bases on litmus paper? 2

19. Differentiate between an open and a closed circuit. 2

20. Differentiate between malleable and ductile. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Differentiate between metals and non-metals on any three properties. 3

22. Describe the activity to show that rusting needs air and water. 3

23. Explain how red cabbage extract can be used as an indicator. 3

24. Why are heating coils made of nichrome? 3

OR

Explain why sodium is kept in kerosene. 3

25. List three changes that are natural and three that are man-made. 3

OR

Why are acid rain and acid spills harmful? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Explain neutralisation with the example of an acid and a base. Write the equation and state two applications. 5

OR

Explain the working of a simple electric circuit and the function of cell, bulb, switch and wires. 5

27. Explain the physical properties of non-metals and compare them with metals in a table-like form. 5

OR

Explain the process of rusting, its conditions and the methods of preventing it. 5

END OF QUESTION PAPER — Total: 50 Marks

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